STAT/MA 41600

In-Class Problem Set #33: November 4, 2026

1. Suppose that a certain virus causes a sickness that lasts for an exponential length of time, with mean 3 days. Suppose that 10 people catch the virus (and the times that they are sick are independent), and we add up the times (given in days) that they are sick $Y = X_1 + \cdots + X_{10}$.

1a. What type of random variable is Y?

1b. What are the parameters of Y?

1c. What is $E(Y)$?

1d. What is $\text{Var}(Y)$?

2. Consider three independent exponential random variables X_1, X_2, X_3, with respective means of 2 minutes, 2.5 minutes, and 3 minutes. Is $Y = X_1 + X_2 + X_3$ a Gamma random variable? Why or why not?

3. Suppose that X_1, X_2, X_3 are independent exponential random variables with $E(X_i) = 1/5$ for each i. Define $Y = X_1 + X_2 + X_3$.

3a. What is $\text{Var}(Y)$?

3b. What is the probability density function of Y?

4a. Revisiting Question **1d**, we notice that $24Y$ is the total number of hours that the 10 people are sick. What is $\text{Var}(24Y)$?

4b. Revisiting Question **3a**, what is $\text{Var}(Y/8)$?

4c. Revisiting Question **3a** again, what is $\text{Var}(6X_1 + 7X_2 + 8X_3)$?

4d. Revisiting Question **3a** again, is $W = 6X_1 + 7X_2 + 8X_3$ a Gamma random variable? Why or why not?

5. There are 4 available types of witches' brew (bubbling, smoky, acidic, and slimy). Upon drinking the potions, each person will begin to feel the effects of the potion in an exponential time with mean 2 minutes (regardless of the type of witches' brew). Suppose that 4 people each simultaneously drink one of the witches' brews, and their times to feel the effects of the potions are independent random variables X_1, X_2, X_3, X_4.

What is the probability density function of $Y = X_1 + X_2 + X_3 + X_4$?

Mind bender (not required). Consider 31 independent Exponential random variables, $X_1, \ldots, X_{31}$, with $E(X_i) = 1/4$ for each i. Find the probability that X_1 is bigger than an even number of the 30 other random variables: $X_2, \ldots, X_{31}$.